



**Namal University, Mianwali**

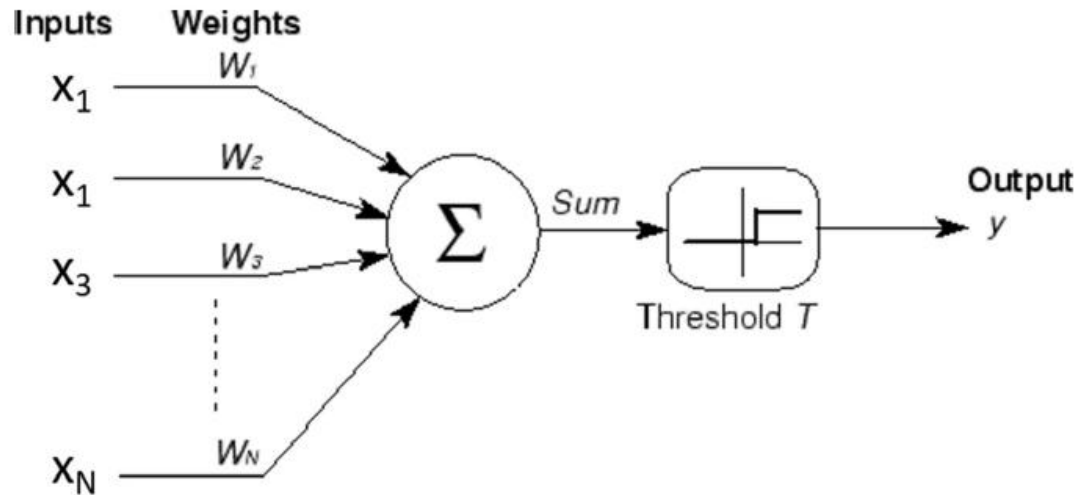
**Department of Computer Science**

# Machine Learning

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# Perceptrons

# Perceptron

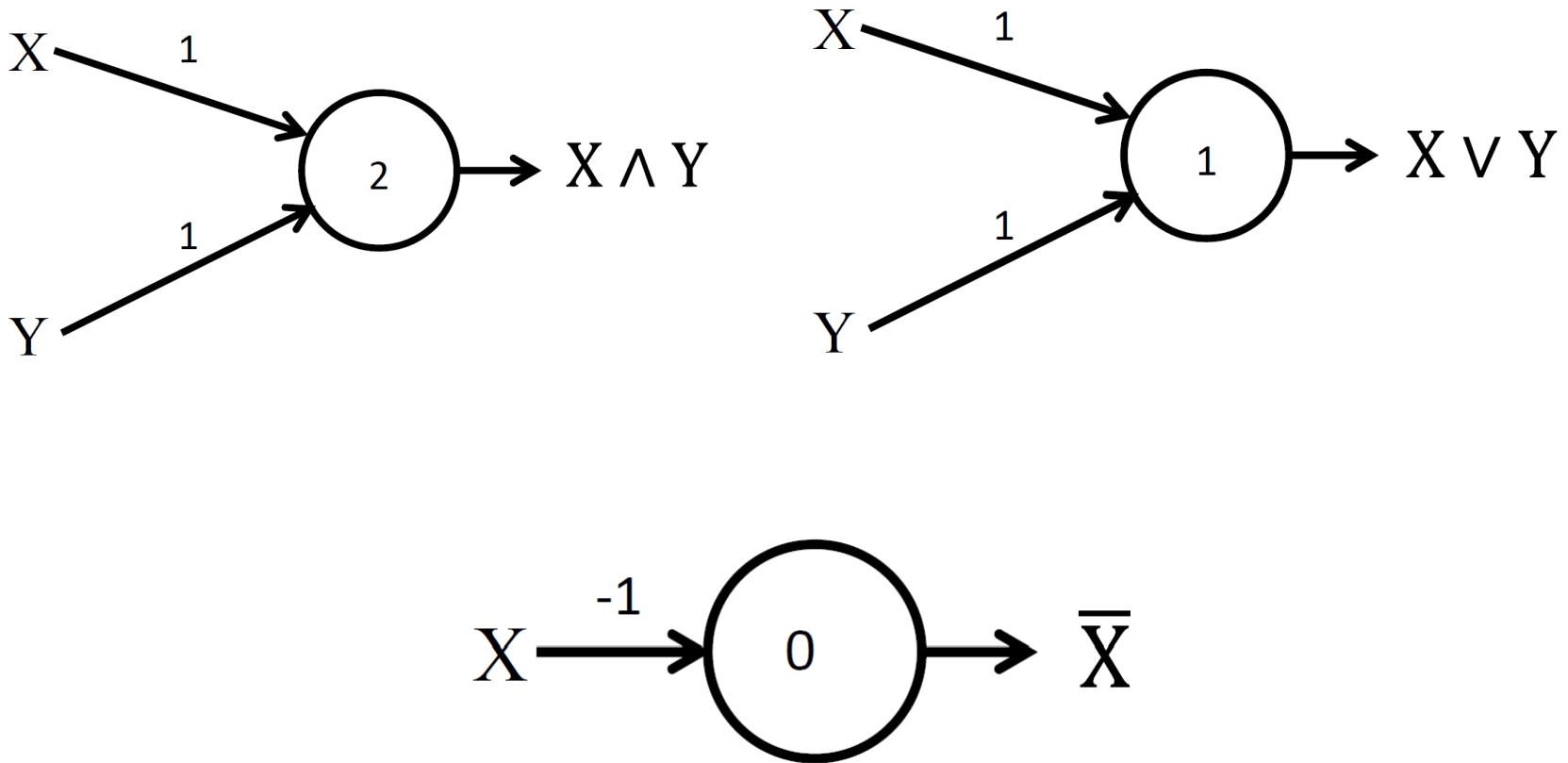


- Number of inputs combine linearly.
- The perceptron fires if the weighted sum is greater than the threshold.

$$Y = \begin{cases} 1 & \text{if } \sum_i w_i x_i - T \geq 0 \\ 0 & \text{else} \end{cases}$$

# Perceptron as Boolean Logic Gates

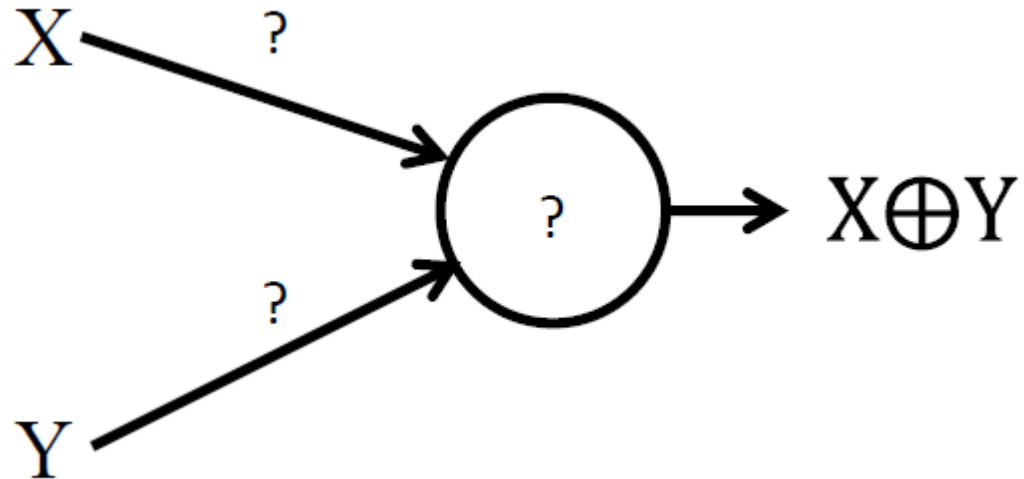
Easily shown to mimic any Boolean gate



# Perceptron as Boolean Logic Gates

Problem?

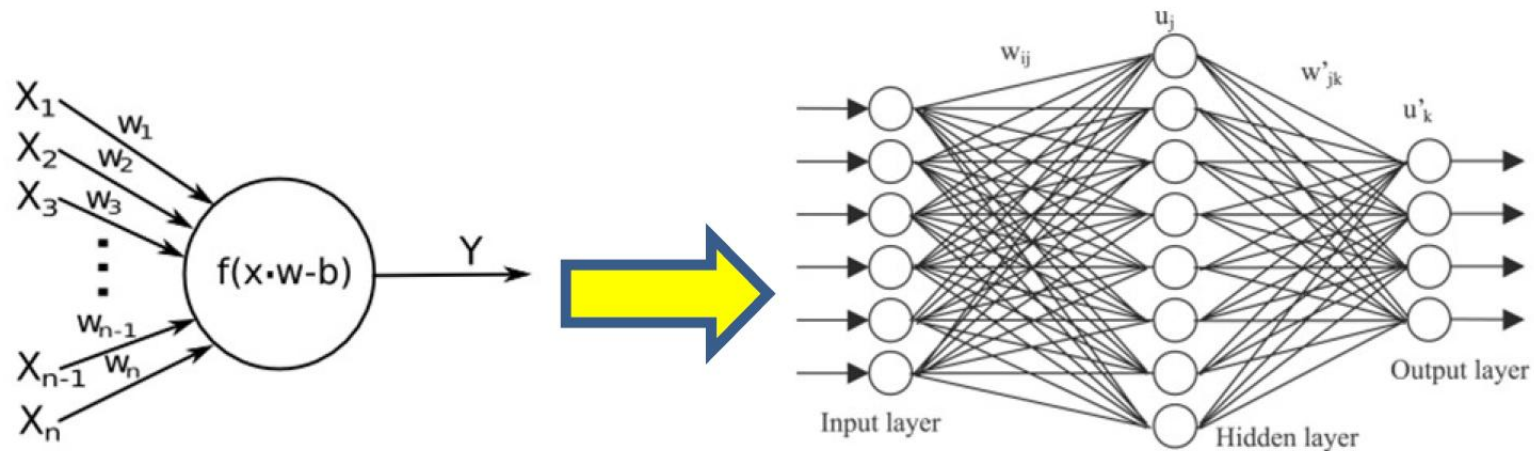
No solution for XOR!



# A Single Perceptron Is not Sufficient

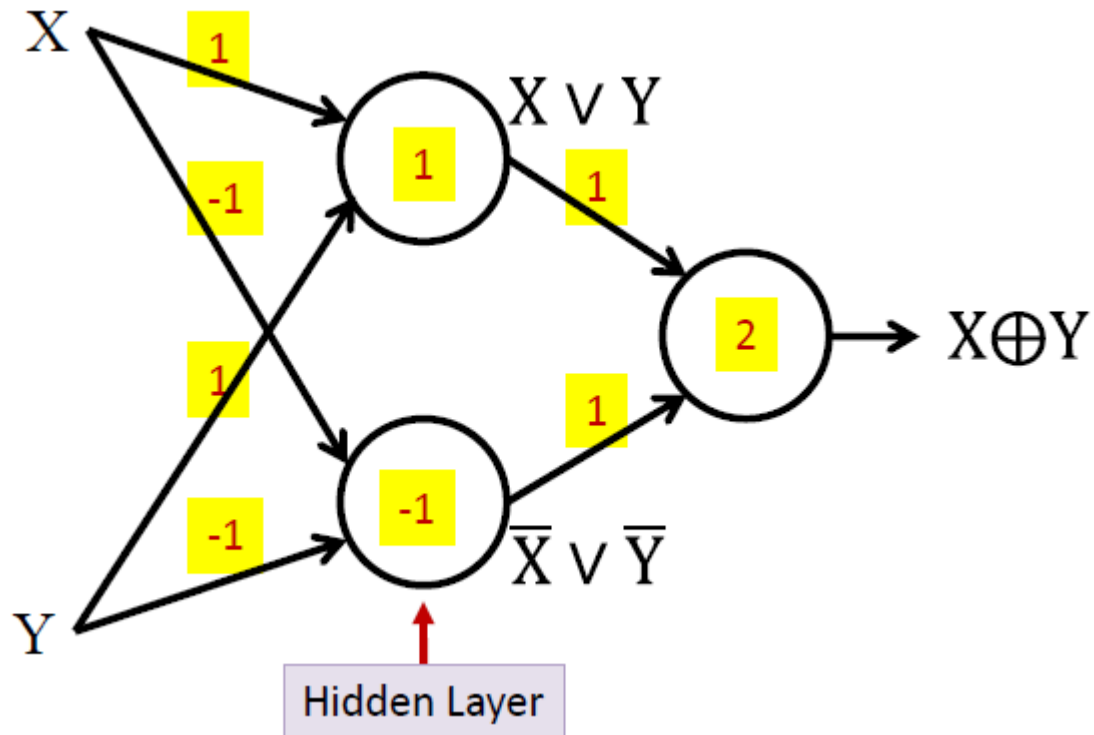
Solution?

Network of Perceptrons are required!



# Solution for XOR

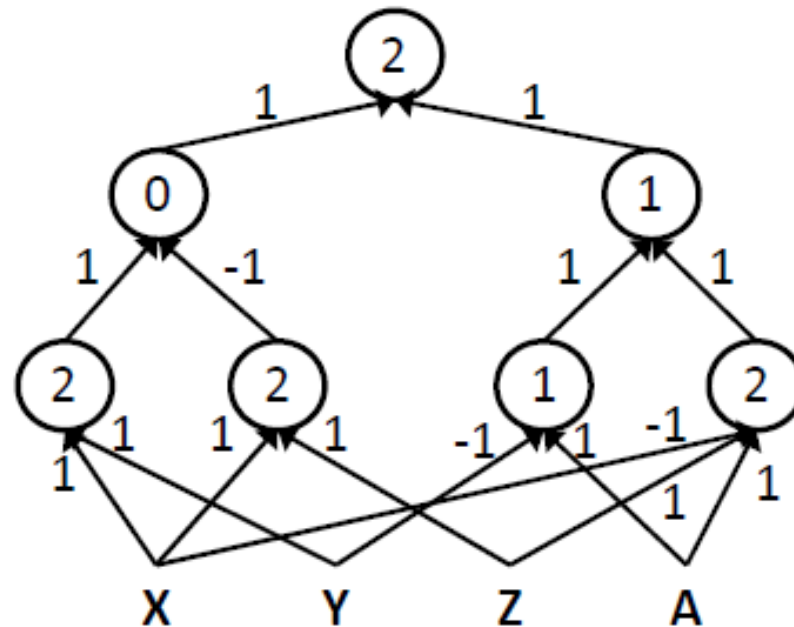
Input, Hidden, and Output layers architecture.



# Any Generic Model

A multi-layer perceptron can implement any arbitrary Boolean function.

$$((A \& \bar{X} \& Z) | (A \& \bar{Y})) \& ((X \& Y) | \overline{(X \& Z)})$$



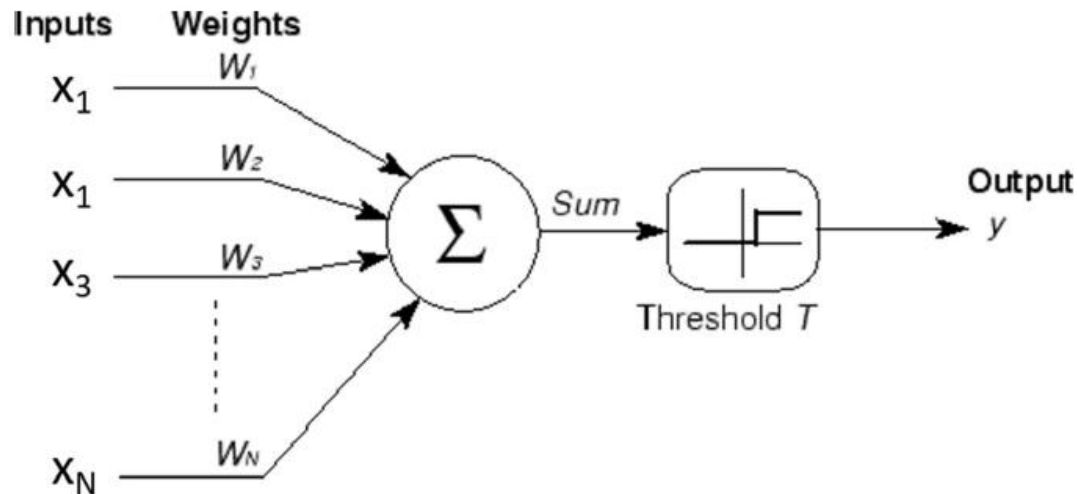


# Perceptrons with Real Inputs

$x_1 \dots x_n$  are real valued.

$w_1 \dots w_n$  are real valued.

Unit fires if the weighted sum of inputs matches or exceeds a threshold.

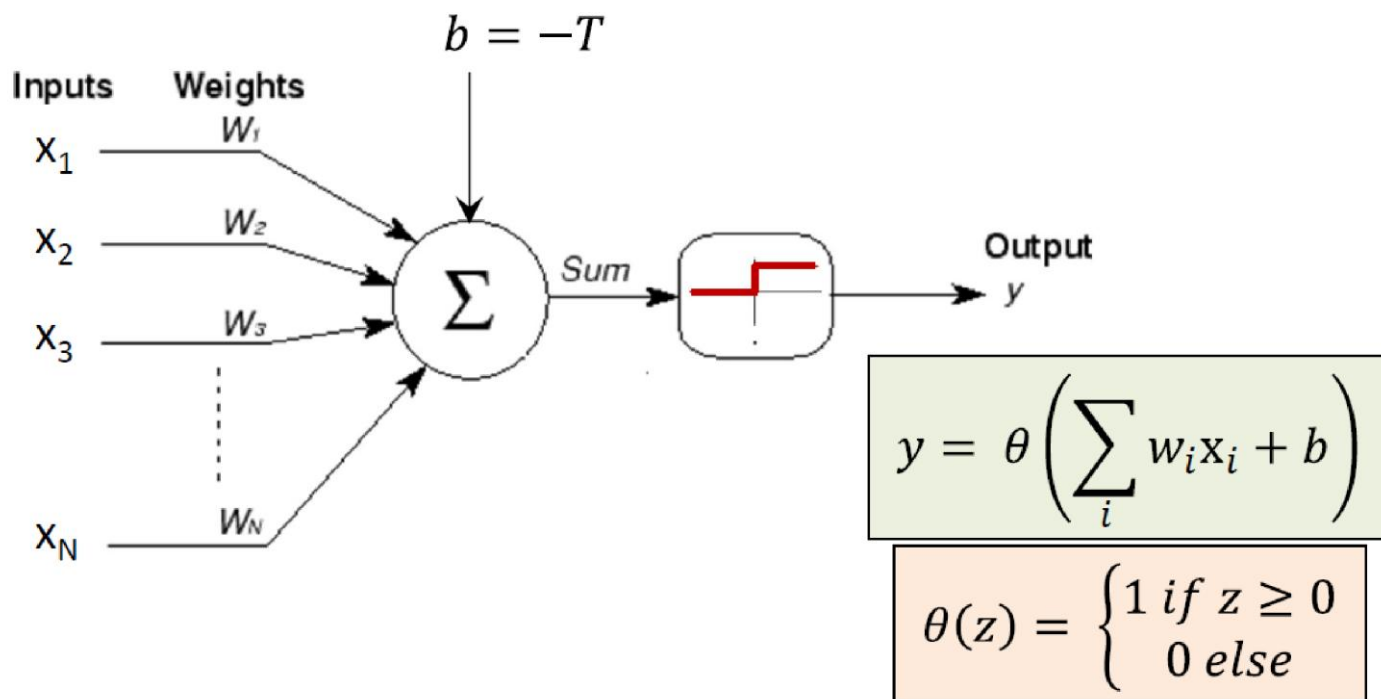


$$y = \begin{cases} 1 & \text{if } \sum_i w_i x_i - T \geq 0 \\ 0 & \text{else} \end{cases}$$

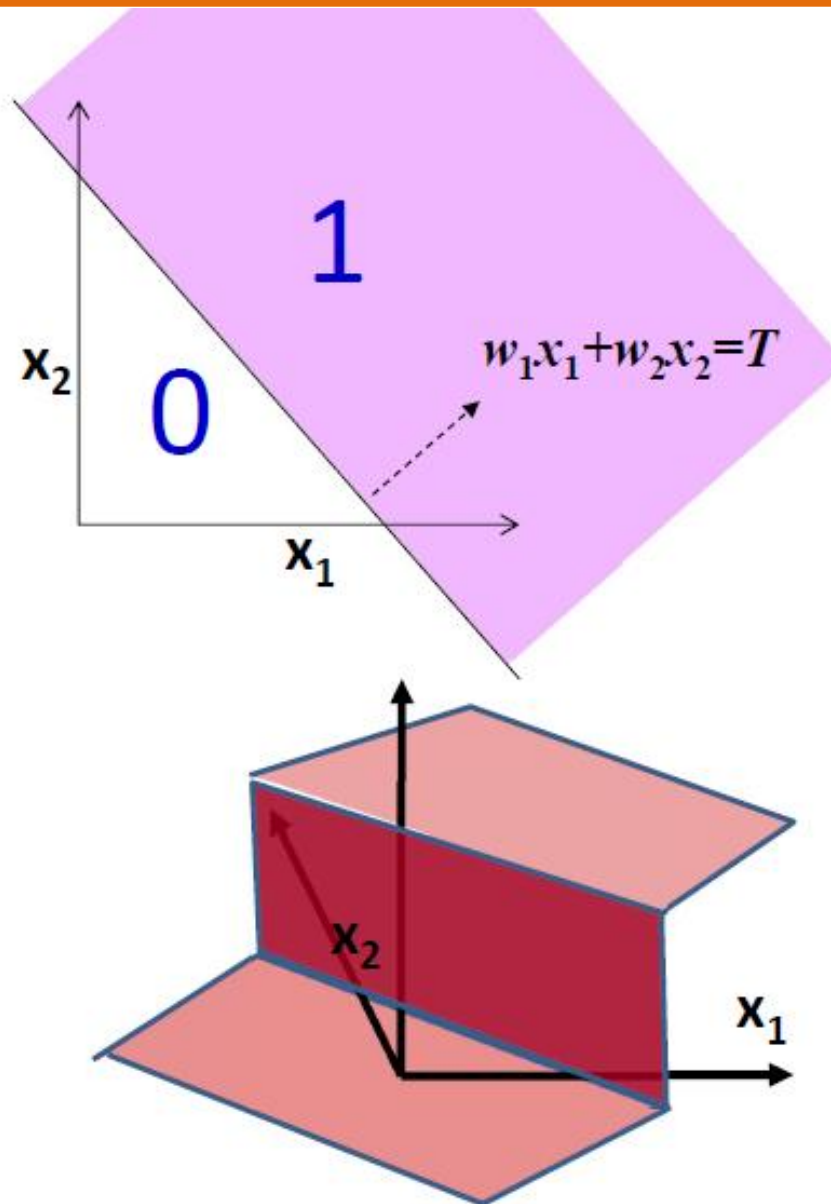
# Perceptrons with Real Inputs

## Alternate Implementation:

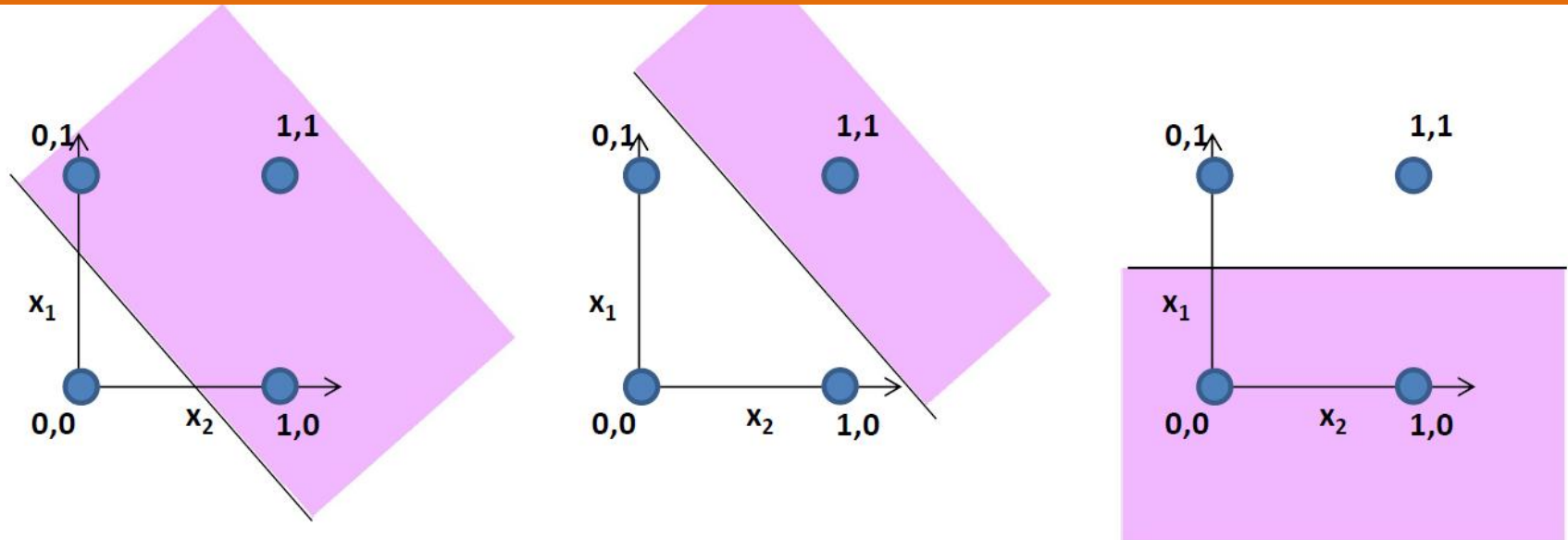
The firing of perceptrons depends upon the weighted sum of inputs plus a bias.



# Perceptrons with Real Inputs



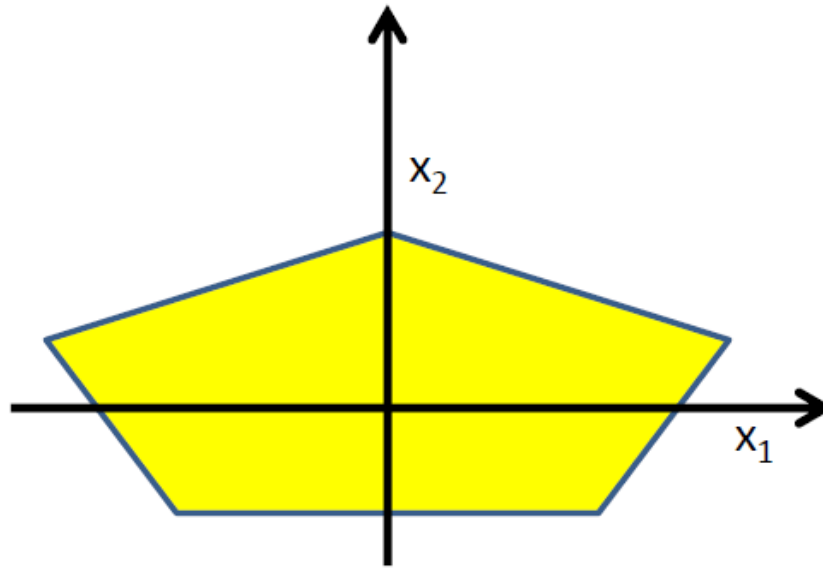
# Perceptrons with Real Inputs



Boolean perceptrons are also linear classifiers

- Purple regions have output 1 in the figures
- What are these functions
- You can see that we cannot compose an XOR with single boundary

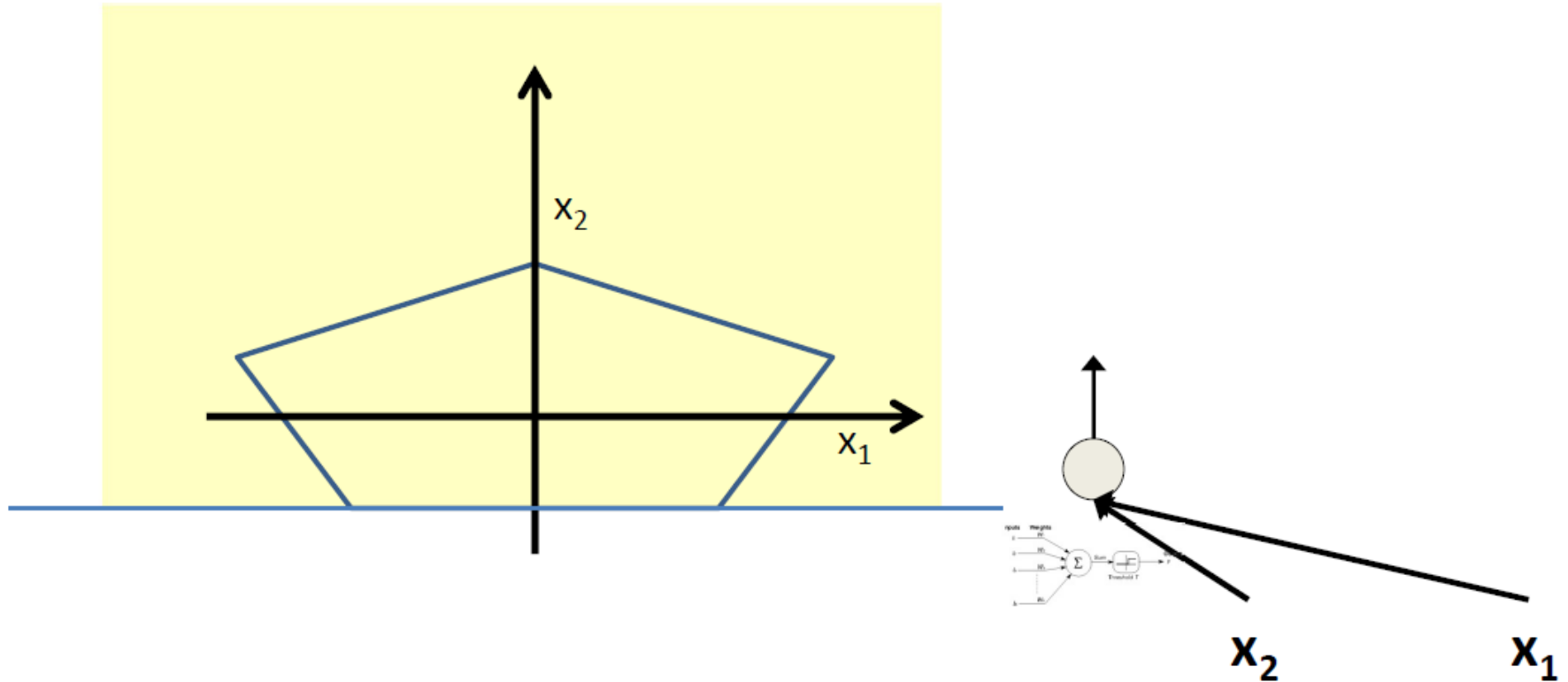
# Arbitrary Complicated Boundary?



Can now be composed into  
"networks" to compute arbitrary  
classification "boundaries"

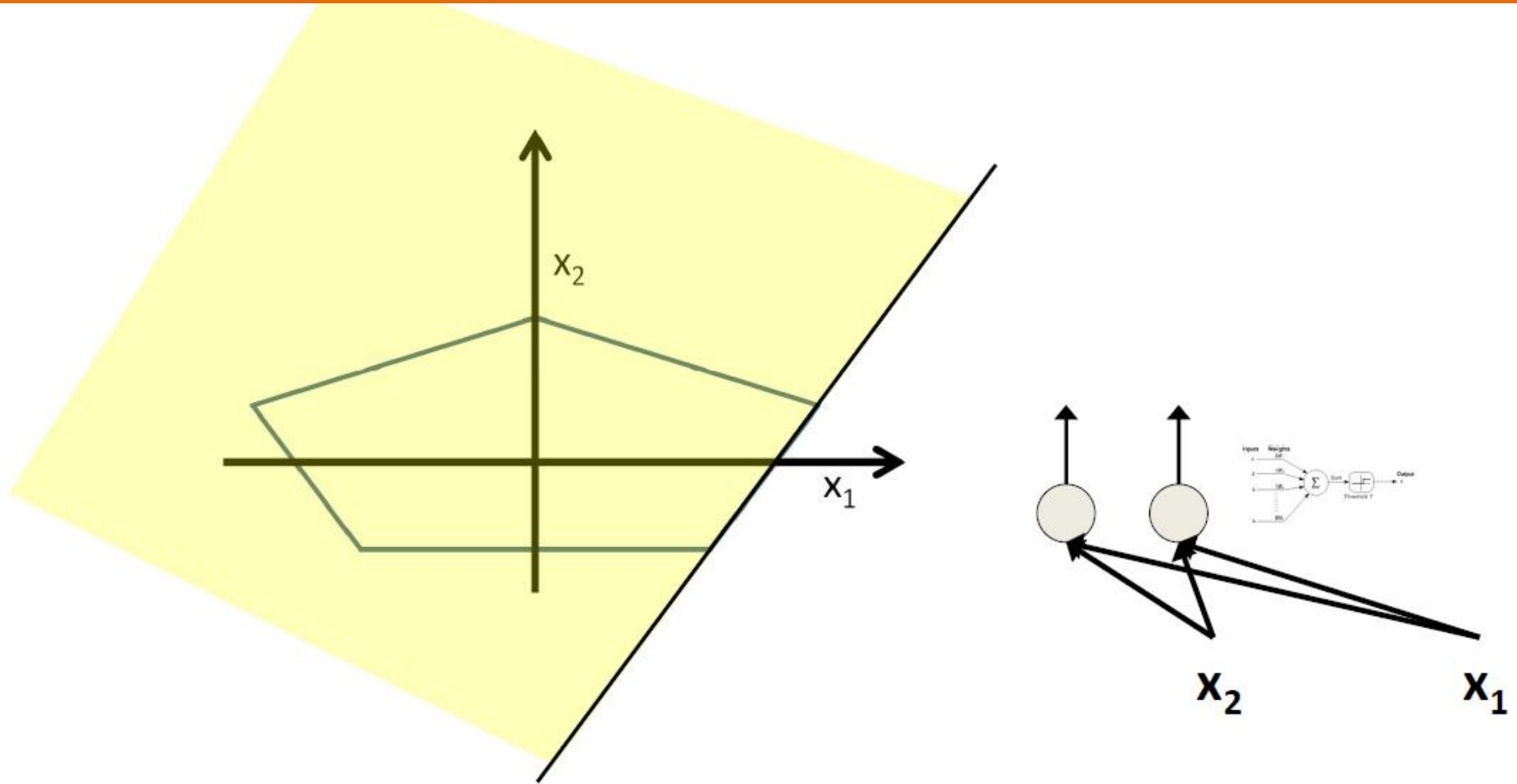
Build a network of units with a single output that fires if  
the input is in the colored area

# Arbitrary Complicated Boundary?



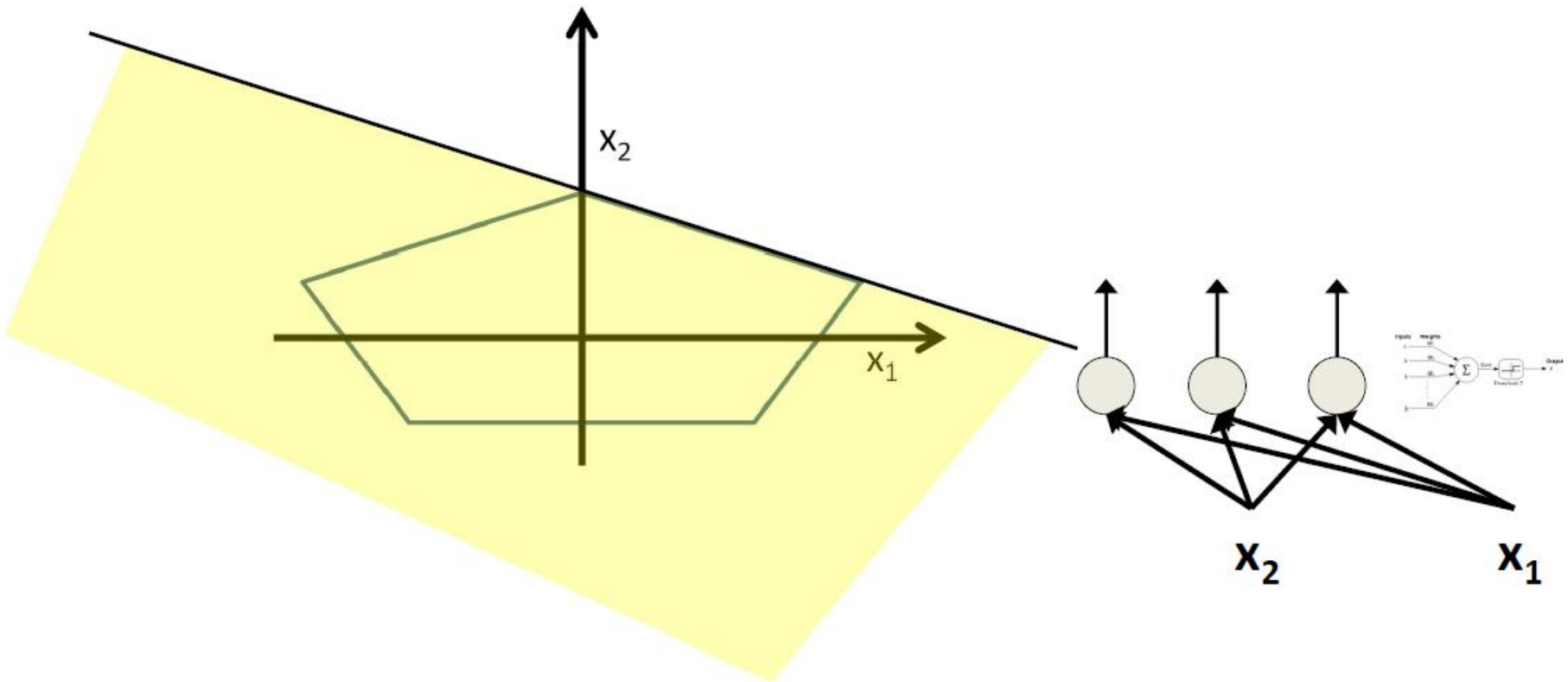
The network will fire in the colored area.

# Arbitrary Complicated Boundary?



The network will fire in the colored area.

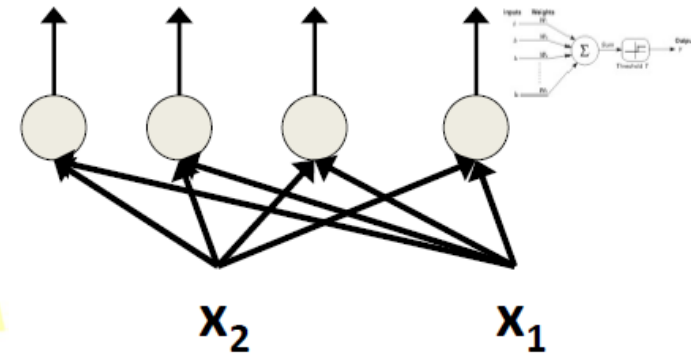
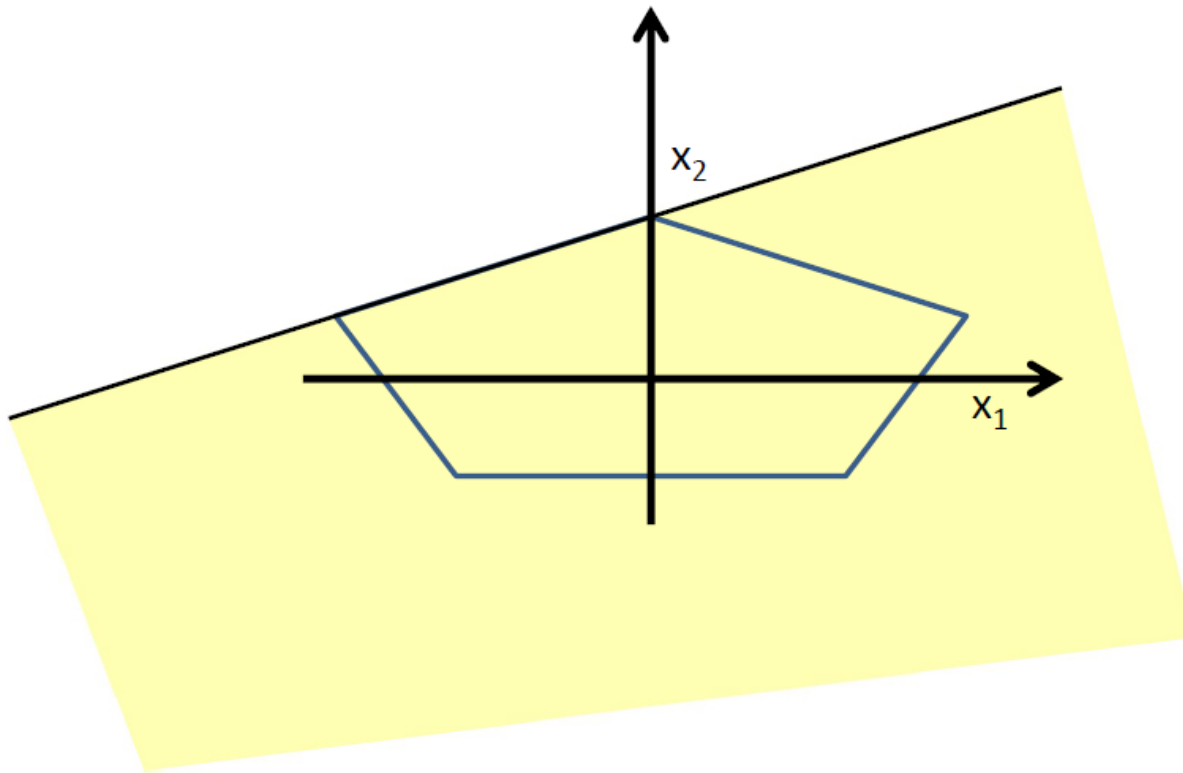
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The network will fire in the colored area.

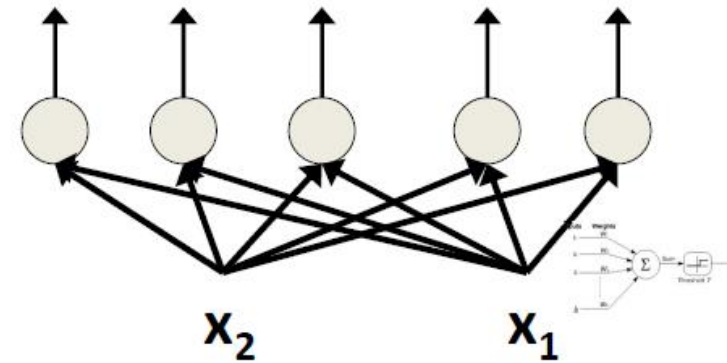
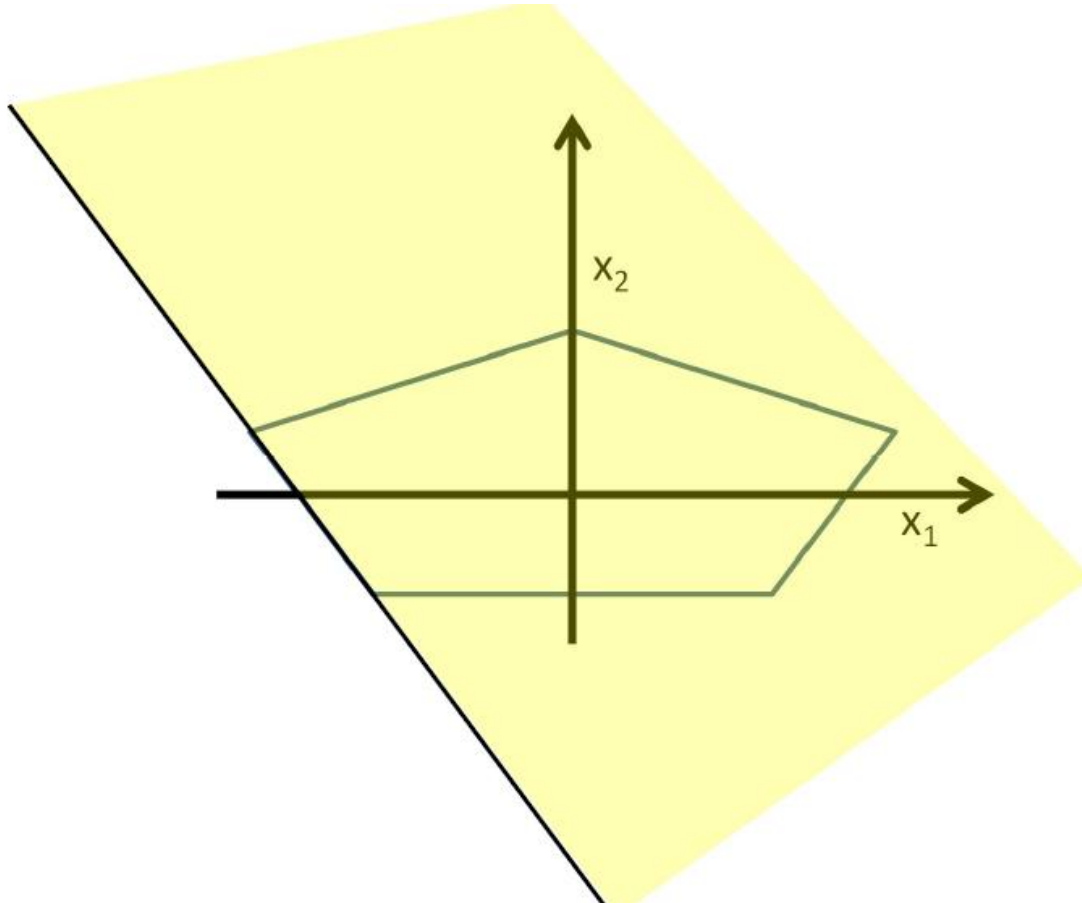


## Arbitrary Complicated Boundary?



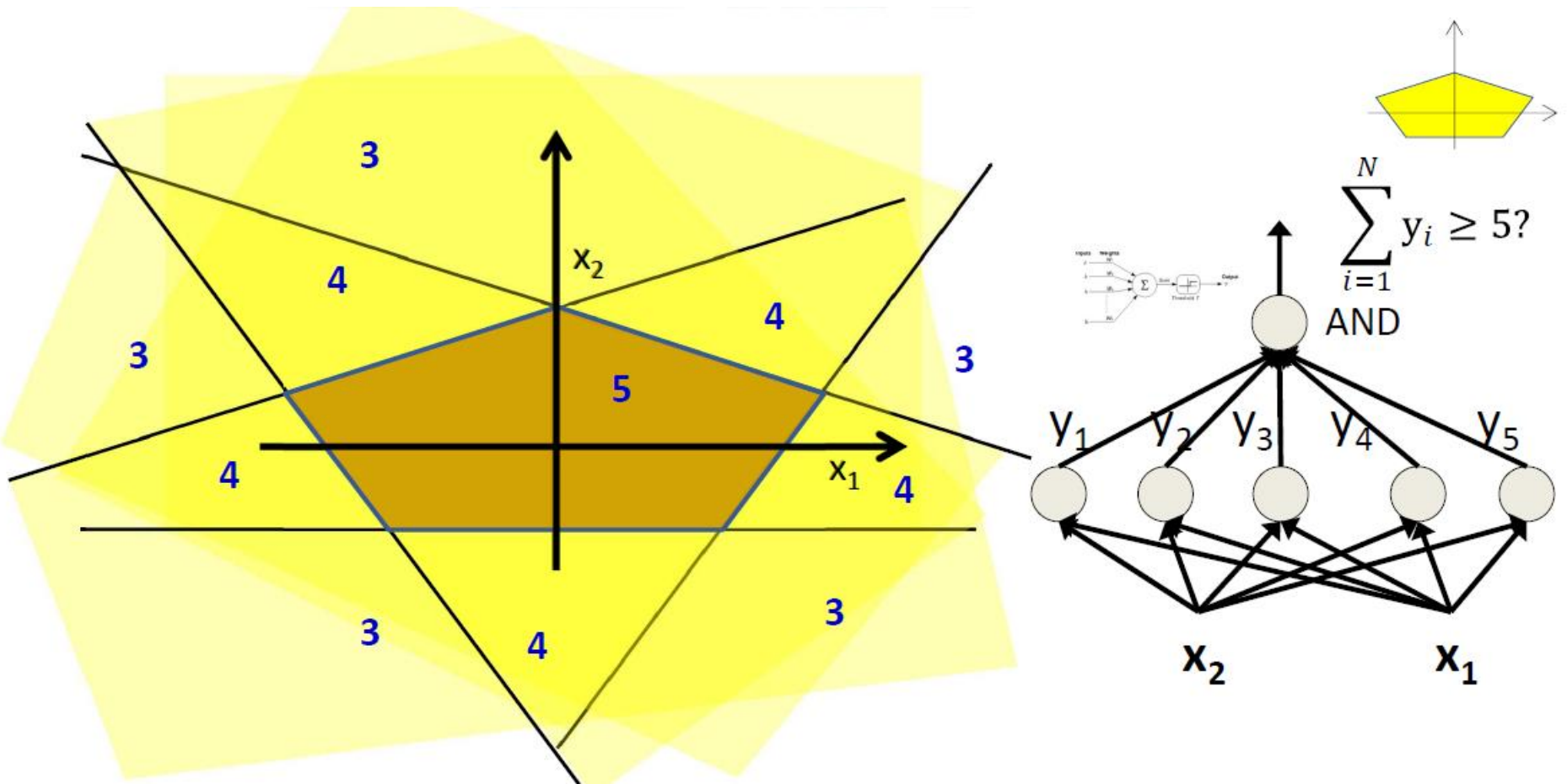
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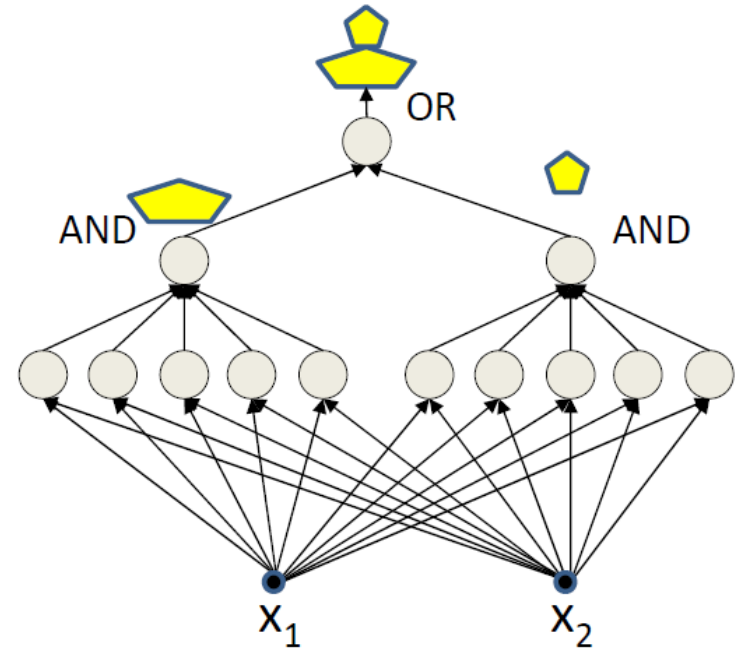
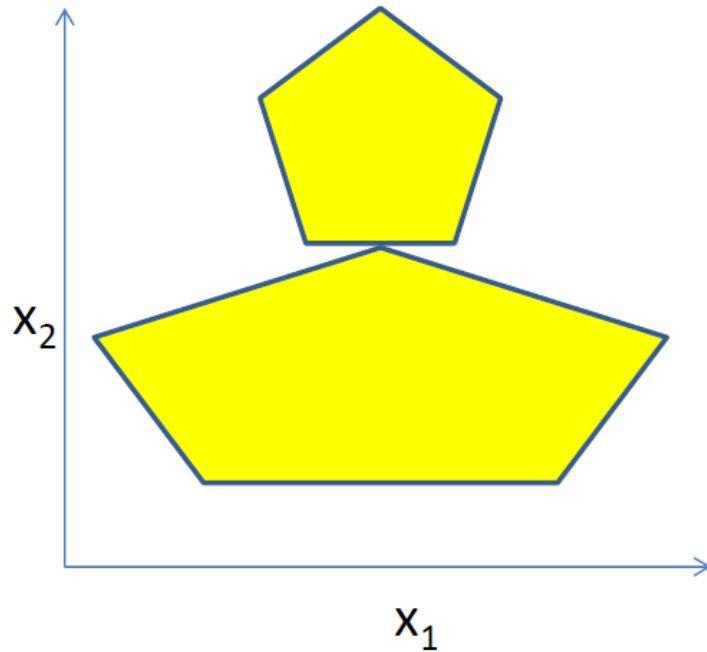
The network will fire in the colored area.

# Arbitrary Complicated Boundary?



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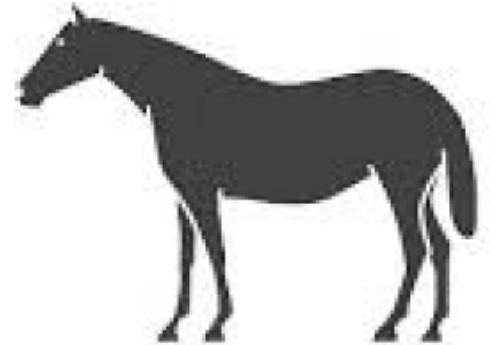
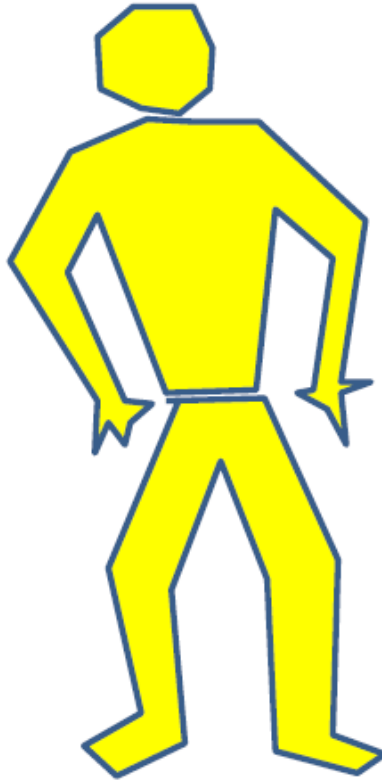
# Multiple Regions?



Possible with a two-layer architecture?

No, third layer is required.

# What about now?



Practically yes.

# Quick Recap

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## MLPs can model any Boolean function

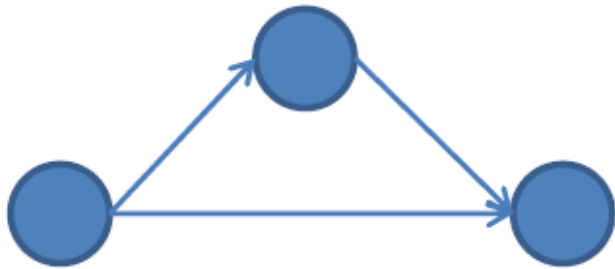
- Individual perceptrons can act as Boolean gates
- Networks of perceptrons are Boolean functions
- MLPs can represent universal Boolean functions

## • MLPs can model any decision boundary

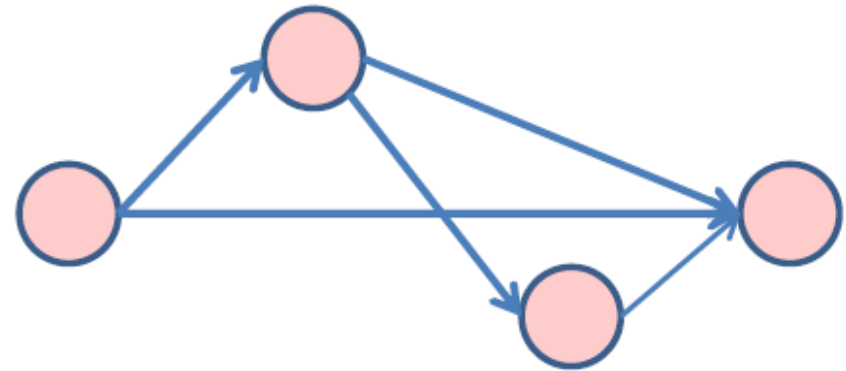
- Individual perceptrons capture linear boundaries
- Complex boundaries can be composed from the linear boundaries
- MLPs can represent arbitrary decision boundaries
- MLPs are universal classifiers

# Defining Depth of a Network

Longest path from the input to the output



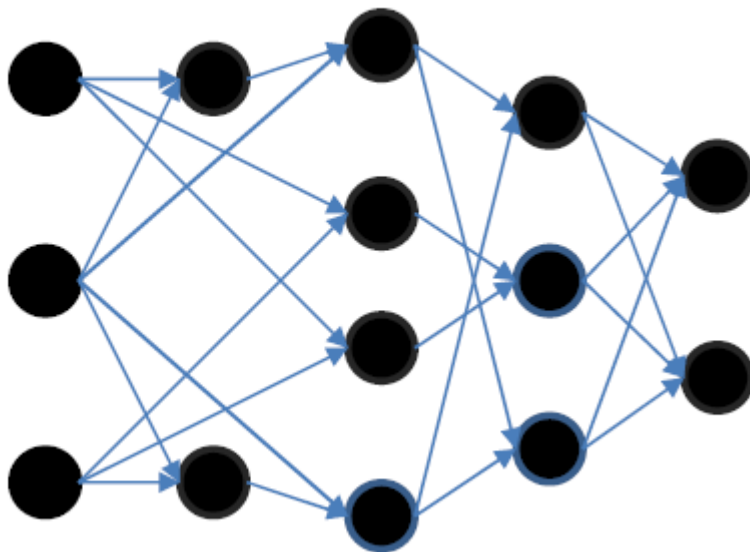
Depth = 2



Depth = 3

# Defining Layer

A layer is the set of neurons that are all at the same depth with respect to the input

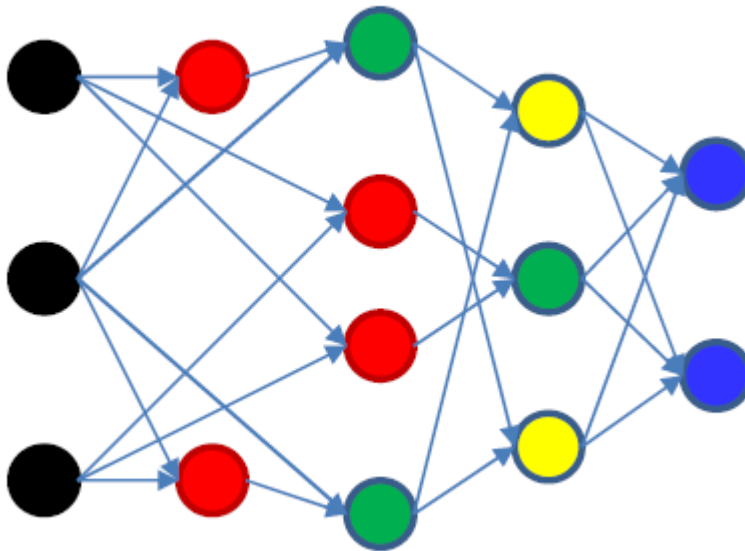


Input:  
Layer 1:  
Layer 2:  
Layer 3:  
Layer 4:



# Defining Layer

A layer is the set of neurons that are all at the same depth with respect to the input



Input: Black  
Layer 1: Red  
Layer 2: Green  
Layer 3: Yellow  
Layer 4: Blue

# How many layers for Boolean MLP

Truth table shows *all* input combinations for which output is 1

Truth Table

| $X_1$ | $X_2$ | $X_3$ | $X_4$ | $X_5$ | $Y$ |
|-------|-------|-------|-------|-------|-----|
| 0     | 0     | 1     | 1     | 0     | 1   |
| 0     | 1     | 0     | 1     | 1     | 1   |
| 0     | 1     | 1     | 0     | 0     | 1   |
| 1     | 0     | 0     | 0     | 1     | 1   |
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$$Y = \bar{X}_1\bar{X}_2X_3X_4\bar{X}_5 + \bar{X}_1X_2\bar{X}_3X_4X_5 + \bar{X}_1X_2X_3\bar{X}_4\bar{X}_5 + X_1\bar{X}_2\bar{X}_3\bar{X}_4X_5 + X_1\bar{X}_2X_3X_4X_5 + X_1X_2\bar{X}_3\bar{X}_4X_5$$

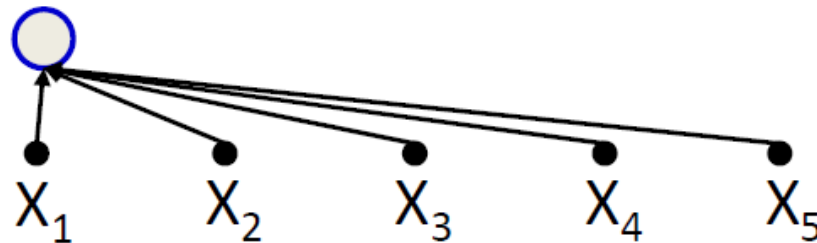
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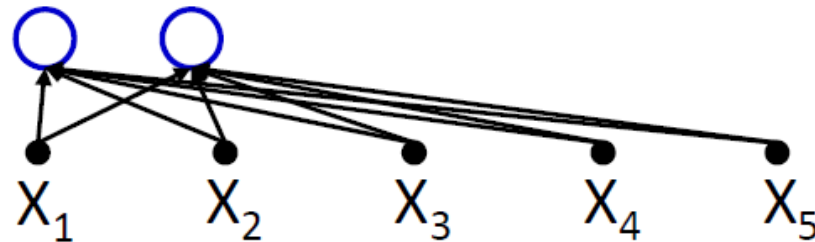
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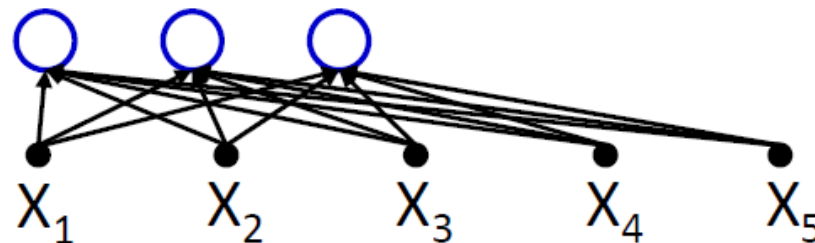
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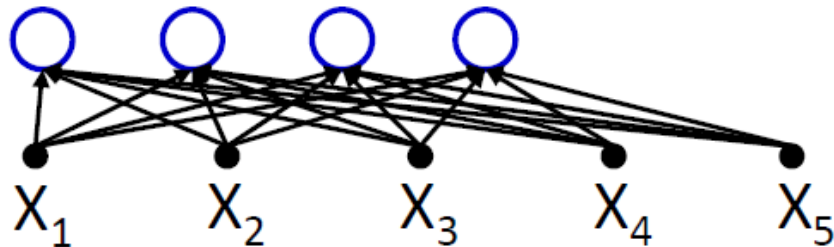
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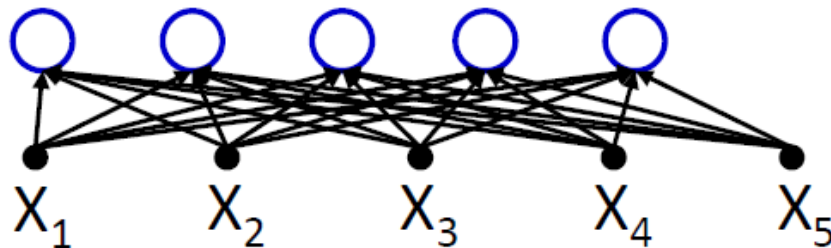
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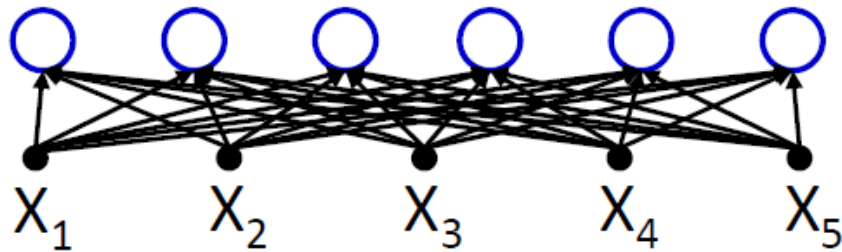
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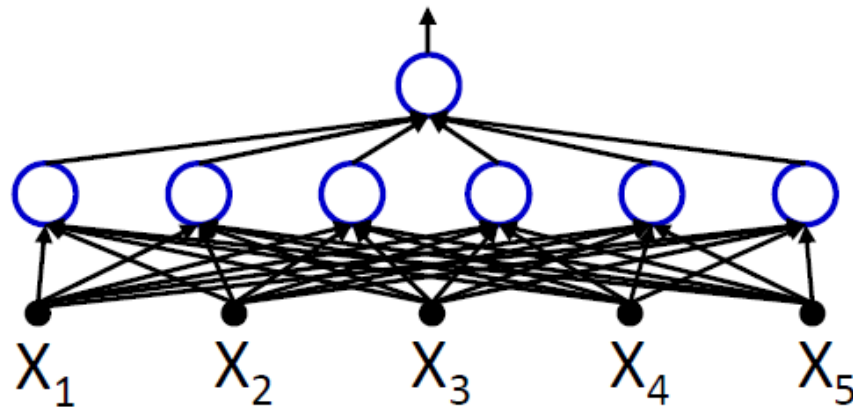
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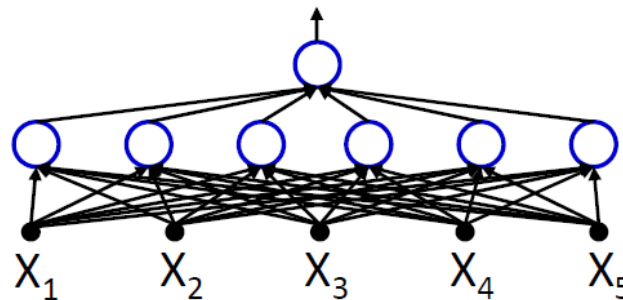
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Any truth table can be expressed in this Canonical Form!

A one-hidden-layer MLP is a Universal Boolean Function

What is the largest number of perceptrons required for the single hidden layer solution for N-input function?

# Reducing the number of Perceptrons

| YZ \ WX | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|
| 00      | 1  | 0  | 0  | 1  |
| 01      | 1  | 1  | 0  | 0  |
| 11      | 1  | 0  | 0  | 0  |
| 10      | 1  | 0  | 0  | 1  |

Basic formula will require 7 terms

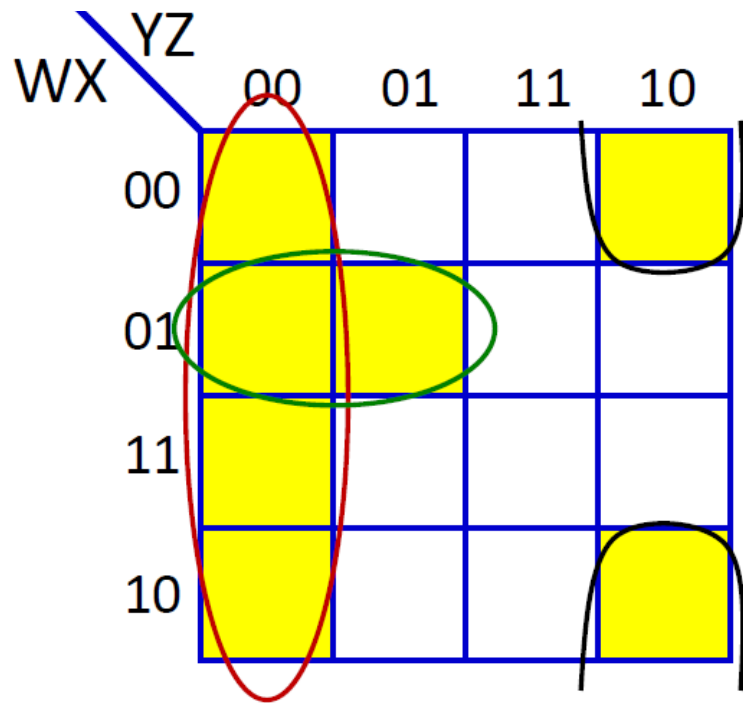
This is a “Karnaugh Map”

It represents a truth table as a grid

Filled boxes represent input combinations for which output is 1; blank boxes have output 0

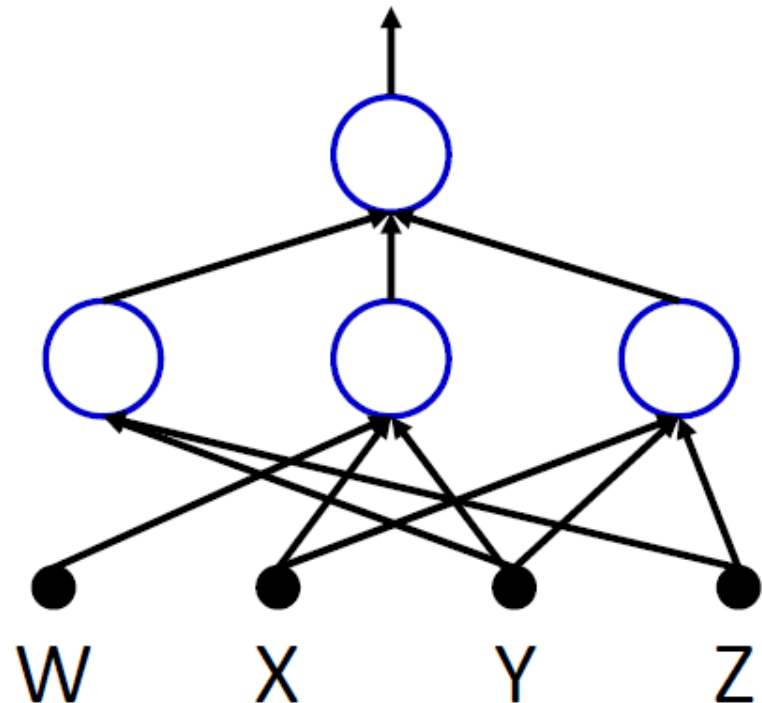
Adjacent boxes can be grouped to reduce the complexity

# Reducing the number of Perceptrons



Boolean network for this function needs only 3 hidden units now.

$$O = \bar{Y}\bar{Z} + \bar{W}X\bar{Y} + \bar{X}YZ$$



# Largest Irreducible Canonical Form

What arrangement of ones and zeros cannot be reduced further?

| WX \ YZ |    |    |    |    |
|---------|----|----|----|----|
|         | 00 | 01 | 11 | 10 |
| 00      |    |    |    |    |
| 01      |    |    |    |    |
| 11      |    |    |    |    |
| 10      |    |    |    |    |

# Largest Irreducible Canonical Form

What arrangement of ones and zeros cannot be reduced further?

| WX \ YZ |  | 00 | 01 | 11 | 10 |
|---------|--|----|----|----|----|
| 00      |  |    |    |    |    |
| 01      |  |    |    |    |    |
| 11      |  |    |    |    |    |
| 10      |  |    |    |    |    |

Red=0  
White=1

# Largest Irreducible Canonical Form

How many neurons for this Boolean function??

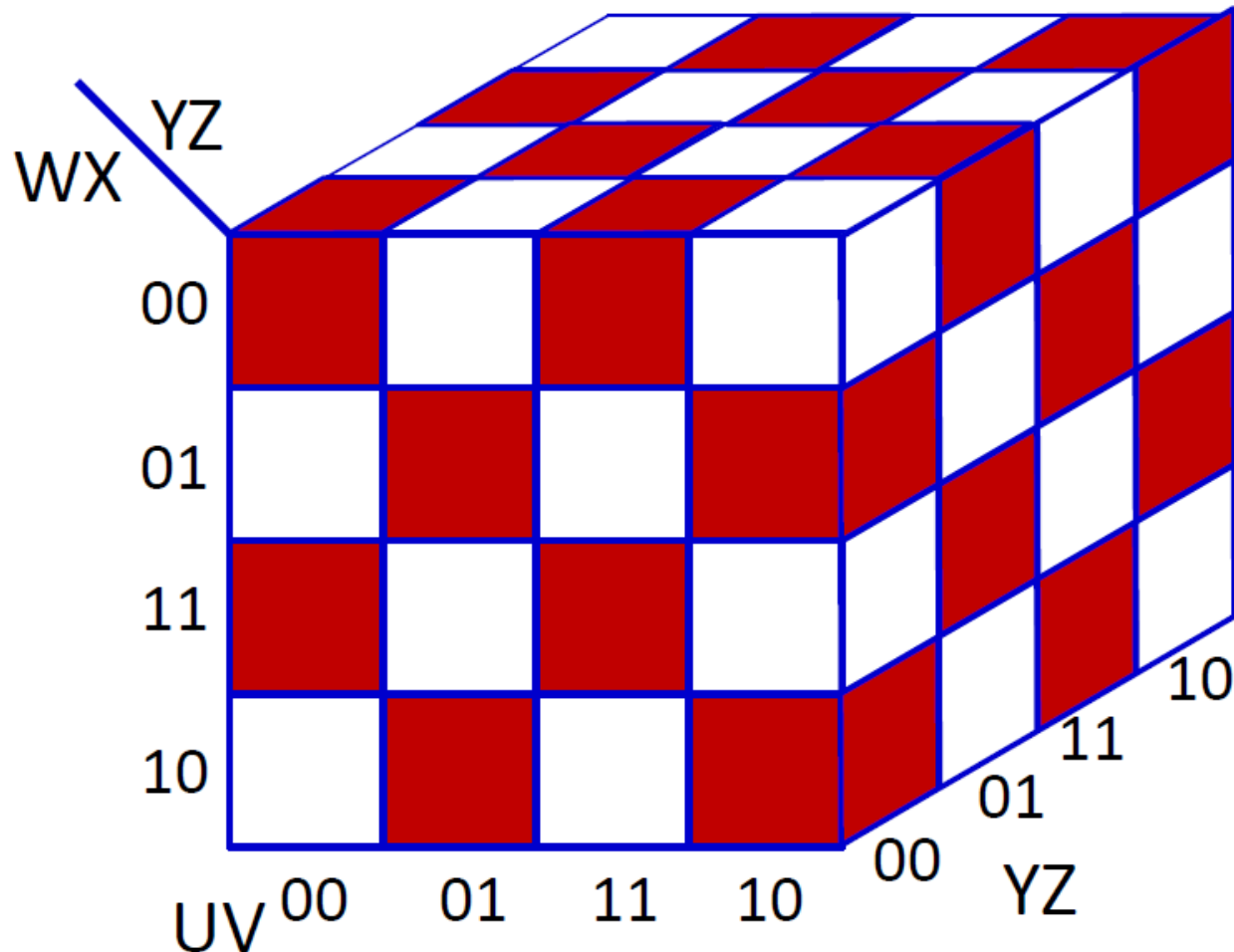
| WX \ YZ |    | 00 | 01 | 11 | 10 |
|---------|----|----|----|----|----|
| WX      | 00 |    |    |    |    |
|         | 01 |    |    |    |    |
|         | 11 |    |    |    |    |
|         | 10 |    |    |    |    |

Red=0  
White=1



# Largest Irreducible Canonical Form

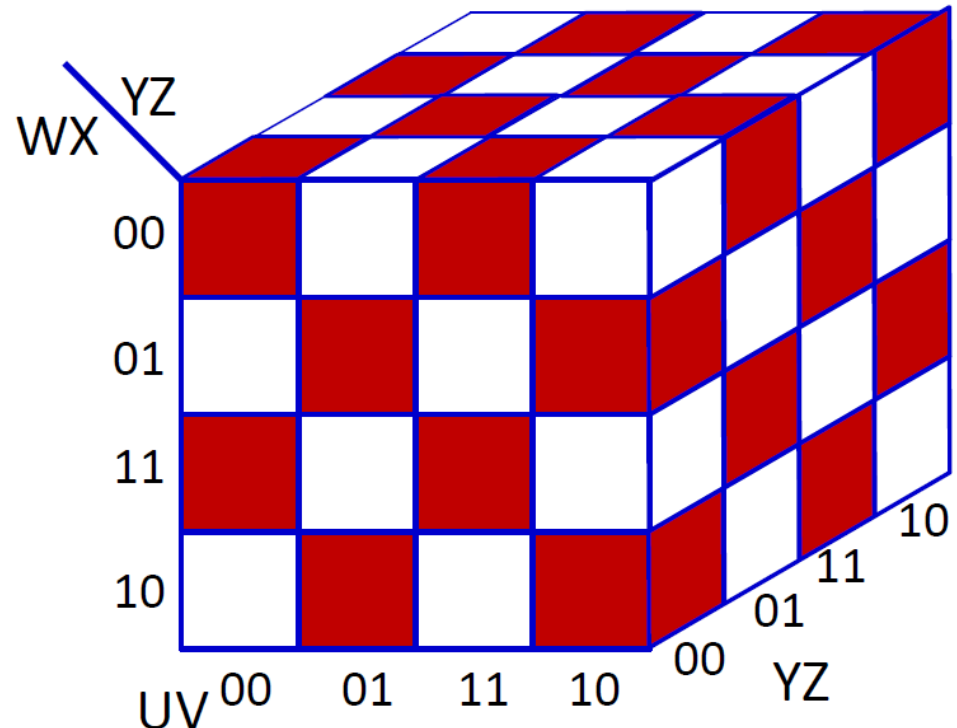
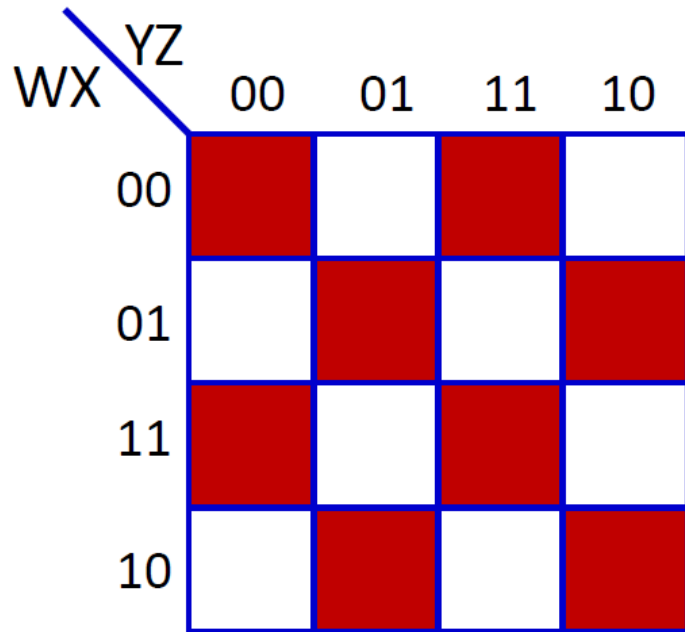
What about this function with 6 variables?



# Largest Irreducible Canonical Form

Can be generalized: Will require  $2^{N-1}$  perceptrons in hidden layer

Exponential in N.



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Thank you